

What Spatial Fine-Tuning Actually Changes: A Multi-Seed Decomposition of Accuracy, Evidence Dependence, and Semantic Consistency in Vision–Language Models

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Abstract

001 *Fine-tuning a vision–language model (VLM) on a spatial*
002 *reasoning task usually improves benchmark accuracy. But*
003 *“better accuracy” is not a single capability: the same adap-*
004 *tation may change how much the model depends on the*
005 *correct image, how coherently it applies relational seman-*
006 *tics, and how it responds to a causal image transformation,*
007 *in different directions and at different rates. We treat or-*
008 *ordinary LoRA fine-tuning on the Visual Spatial Reasoning*
009 *(VSR) task as a controlled intervention and measure, per*
010 *training seed and across two backbone families, a vector*
011 *of capability changes: benchmark gain ΔA , correct-image*
012 *dependence ΔG (the normal-minus-shuffle accuracy gap,*
013 *relative to zero-shot), and semantic pair consistency ΔC*
014 *(relcomp/facing-antonym flip-law compliance), together with*
015 *Tier-C transformation behavior under global horizontal re-*
016 *flexion (transformed accuracy, pair consistency, joint cor-*
017 *rectness). Across 3 fresh training seeds per family (plus the*
018 *committed seed-0), ΔA is positive for every seed on both*
019 *backbones (7B: $+0.0506 \pm 0.0046$; 2B: $+0.0316 \pm 0.0009$)*
020 *and ΔG is positive for every seed (7B: $+0.0440 \pm 0.0054$;*
021 *2B: $+0.0316 \pm 0.0026$). Fresh-seed semantic pair consis-*
022 *tency stays within 0.05 of the legacy General checkpoint*
023 *in both families, and all fresh 2B answer-update rates ex-*
024 *ceed the 2B zero-shot rate. A post-confirmatory exten-*
025 *sion on the contemporary Qwen3-VL-8B backbone repro-*
026 *duces the primary $\Delta A/\Delta G$ pattern ($+0.032/+0.036$) and*
027 *the transformed-accuracy gain ($+0.045$). We conclude that*
028 *ordinary spatial fine-tuning is not a scalar capability update:*
029 *benchmark accuracy, correct-image dependence, and seman-*
030 *tic consistency can be described—and must be reported—as*
031 *a jointly measured vector whose components replicate across*
032 *seeds and families.*

1. Introduction

Spatial reasoning is central to multimodal systems that operate in physical environments: a useful VLM must determine whether one object is to the left of another, above it, in front of it, or facing it, and it must do so using the image—not a language prior. A growing line of work shows that benchmark accuracy can diverge from genuine visual grounding: models can answer correctly while depending weakly on the image [2, 6, 7], and consistency across conditions can coexist with error [1]. Prior work has also introduced grounding-aware training objectives that explicitly reward visual dependence [5].

What has not been examined is whether *ordinary* spatial task fine-tuning—the standard LoRA recipe with no grounding-aware objective—moves benchmark accuracy, correct-image dependence, and semantic pair consistency as a jointly measured vector of capability changes, and whether those changes replicate across training seeds and backbone families. This is the gap we address.

We use Visual Spatial Reasoning (VSR) [3] as the task. VSR contains natural-image statements about object-relative relations (left/right, above/below, facing, containment, and others) paired with true/false labels. Following the frozen protocol of the underlying study (documented in the repository decision log), we evaluate every checkpoint under a corrected battery of six conditions: *normal* (the frozen prompt and image), *shuffle* (frozen derangement of the image), *relcomp* (strict complement pairs: language changes, expected label flips), *facingcomp* (facing-antonym flip-law diagnostic), and the Tier-C pair *hflip_flip* / *hflip_invariant* (global horizontal reflection; language fixed; left/right truth must flip, vertical/depth truth must stay).

We train three fresh General-LoRA seeds per backbone (7B Qwen2-VL-7B and 2B SmolVLM2-2B) using the identical recipe as the committed seed-0 run, then measure, per seed:

- ΔA (**benchmark gain**): normal accuracy after tuning minus zero-shot normal accuracy;

- ΔG (**correct-image dependence**): the change in the normal-minus-shuffle accuracy gap relative to zero-shot;
- ΔC (**semantic pair consistency**): relcomp/facingcomp flip-law compliance and its seed stability;
- **Tier-C transformation behavior**: three separate quantities under global reflection—transformed accuracy ($A_{\text{transform}}$), pair consistency (C_{pair}), and joint correctness (both_correct)—following the frozen metric definitions of the underlying study.

Every headline number in this paper was recomputed independently from raw prediction artifacts by a standalone audit script (`scripts/audit_paper2_numbers_independent.py`) explicitly that imports no analyzer module; the audit passed with zero discrepancies and zero failed claims. We report means and standard deviations across the three fresh seeds; seeds are independent training draws, not ordered stages.

Our primary finding is that the adaptation decomposes cleanly and reproducibly. ΔA is positive for all six fresh seeds across both families; ΔG is positive for all six; semantic pair consistency stays within 0.05 of the legacy General checkpoint for every fresh seed; and all fresh 2B Tier-C answer-update rates exceed the 2B zero-shot rate. A post-confirmatory extension on the contemporary Qwen3-VL-8B backbone reproduces the primary $\Delta A/\Delta G$ pattern and the transformed-accuracy gain, and is explicitly labeled exploratory rather than preregistered.

We make three contributions:

- **A controlled multi-seed, cross-backbone decomposition of ordinary spatial fine-tuning** into benchmark gain, correct-image dependence, and semantic pair consistency, with all quantities recomputed independently and frozen before manuscript mode.
- **A seed-robust result set**: every component of the vector replicates across three fresh training seeds on two backbone families, closing the single-checkpoint-anecdote criticism.
- **A scoped contemporary-backbone extension** (Qwen3-VL-8B) that reproduces the primary adaptation pattern, honestly labeled as post-confirmatory external validation.

2. Related Work

Accuracy versus visual grounding. A central and rapidly growing observation is that benchmark accuracy can overstate genuine visual dependence. *Beyond Accuracy* (Zafar et al., 2026) [6] evaluates medical VLMs with real, blank, and shuffled images and defines the Visual Reliance Score (VRS) as $A(\text{real}) - A(\text{shuffle})$; they find that RLVR can improve accuracy while degrading visual grounding. Our *correct-image dependence* (the normal-minus-shuffle gap, G) is the same construction; we do not claim it as novel, and we report the zero-shot-relative change ΔG as the adaptation quantity. *Seeing without Looking* (Lan et al., 2026) [2] applies global

and localized visual interventions to show that predictions can remain robust after evidence removal; it audits pretrained models rather than a fine-tuning transition. *VisualFLIP* (Zhu et al., 2026) [7] introduces a paired benchmark where the question stays fixed and a minimal task-critical edit deterministically flips the gold answer, with pair accuracy and Collapse Rate as metrics; its dataset and harness are not yet publicly released, and we do not claim to reproduce its protocol. Our Tier-C horizontal reflection is a global image intervention, not a minimal local edit; we report our pair metrics as *collapse-style paired answer-update metrics following VisualFLIP*, with the intervention difference stated

Consistency versus grounding. *Consistent Yet Wrong* (Bhat and Yamasaki, 2026) [1] introduces ViewDiag and shows that multi-view consistency can coexist with substantial error, indicating prior-driven collapse rather than evidence-sensitive reasoning. Their dissociation between consistency and grounding is the static analogue of our ΔC -versus- ΔG decomposition, but they do not study a fine-tuning transition. *VisualFLIP*'s collapse rate is a similar paired-consistency notion for minimal-edit pairs.

Grounding-aware training objectives. *CORAL* (Zafar et al., 2026) [5] fine-tunes Qwen2.5-VL-7B with a Contrastive Grounding Objective that penalizes answer invariance under hard-negative image swaps, improving accuracy and reducing visual hallucination in medical VQA. *Med-OPD* (Qian et al., 2026) [4] adds evidence-aware on-policy distillation for medical VLMs. These works *design* grounding-aware objectives; we study what *ordinary* fine-tuning already changes, without modifying the objective.

Positioning. No prior work, to our knowledge, treats ordinary spatial task fine-tuning as a controlled intervention and measures ΔA , ΔG , and ΔC (plus Tier-C transformation behavior) jointly, per training seed, across two backbone families, with a contemporary-backbone extension. Our novelty claim is precisely this joint, seed-robust decomposition—not the normal-minus-shuffle gap, not “accuracy differs from grounding,” and not a new training objective.

3. Protocol

Task and data. We use Visual Spatial Reasoning (VSR) [3], test split ($n = 2195$ rows). Statements are true/false sentences about object-relative relations; labels are frozen. The frozen prompt template, generation settings (greedy, $\text{max_new_tokens} = 5$), image preprocessing (uniform 392 px long-side cap), and cache keys (md5 of the image URL) are identical to the underlying committed protocol (documented in the repository decision log and protocol

171 freeze documents).

172 **Evaluation battery (corrected, frozen).** Following the
173 battery-drift correction in the decision log, the seed campaign
174 is evaluated with the already-committed legacy grounding
175 drivers, not the drifted heavy battery. The corrected battery
176 has six conditions with canonical frozen subset sizes:

177	condition	n
	normal / shuffle / blank / text_only	2195
	relcomp (strict complements)	666
178	facingcomp (facing-antonym)	103
	hflip_flip (global reflection, flip-expected)	245
	hflip_invariant (global reflection, expected-invariant)	421

179 The shuffle condition uses the frozen cycle-derangement
180 mapping (seed 20260810); identity and eligibility come
181 from frozen *eligible-ids* manifests. Attention is eager for
182 Qwen2-VL-7B and sdpa for SmolVLM2-2B, matching every
183 committed legacy run.

184 **Training recipe (verbatim seed-0 recipe).** Each fresh
185 seed is a General-LoRA run on the general training manifest
186 (2000 rows; 95/5 split frozen at seed 42) with the seed-0
187 recipe: epochs 2, lr 10^{-4} , LoRA $r=8$, $\alpha=16$, dropout 0.05,
188 bf16, gradient checkpointing. The only intentional differ-
189 ence from seed-0 is the explicit per-run seed (101/202/303)
190 applied to torch/cuda/numpy at driver start. Three seeds per
191 backbone were trained on A6000 GPUs.

192 **Fresh-seed analysis.** Seeds are independent training
193 draws, not ordered stages; we report per-seed values, means,
194 and standard deviations, and never “monotonic across seeds.”
195 All comparisons are within-backbone; preprocessing is iden-
196 tical within a backbone.

197 **Independent numerical audit.** Every headline quan-
198 tity in this paper was recomputed by a standalone script
199 (`scripts/audit_paper2_numbers_independent.py`)
200 that imports no analyzer module and reads only raw pre-
201 diction CSVs/JSONL, frozen protocol manifests, and
202 committed analysis JSONs as comparison targets. The audit
203 hard-fails on any deterministic-proportion discrepancy above
204 10^{-6} , any row-count/ID mismatch, and any terminology
205 mismatch. It passed with zero issues, and the committed
206 seed-0 tier-a values were reproduced byte-identically (7B
207 normal 0.82414579; 2B normal 0.76492027; absolute
208 difference 0.0).

209 4. Results

210 All values below are audit-sourced (Step-1 independent audit,
211 PASS). Seeds are reported per-draw; means/SDs are over the
212 three fresh seeds.

family	checkpoint	ΔA	G	ΔG
Qwen2-VL-7B	seed-0	+0.0542	0.3522	+0.0456
Qwen2-VL-7B	seedA	+0.0547	0.3544	+0.0478
Qwen2-VL-7B	seedB	+0.0456	0.3444	+0.0378
Qwen2-VL-7B	seedC	+0.0515	0.3531	+0.0465
SmolVLM2-2B	seed-0	+0.0287	0.2975	+0.0305
SmolVLM2-2B	seedA	+0.0305	0.3007	+0.0337
SmolVLM2-2B	seedB	+0.0323	0.2957	+0.0287
SmolVLM2-2B	seedC	+0.0319	0.2993	+0.0323
Qwen3-VL-8B	tuned	+0.0323	0.3827	+0.0355

Table 1. Headline adaptation quantities. $G = A_{\text{normal}} - A_{\text{shuffle}}$; ΔG is relative to zero-shot. All values from the independent audit.

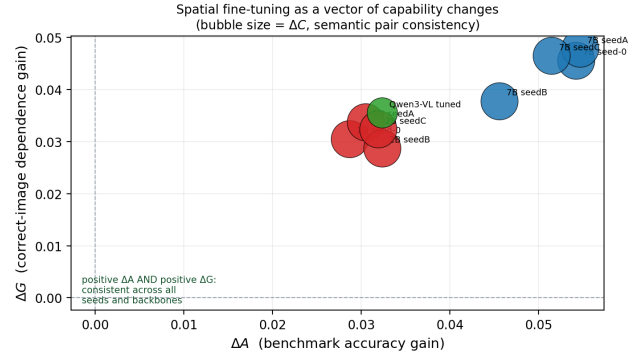


Figure 1. Conceptual decomposition: ΔA (x) versus ΔG (y) for all nine checkpoints (7B seed-0 + three seeds, 2B seed-0 + three seeds, Qwen3-VL-8B tuned); bubble size encodes ΔC (semantic pair consistency change). All points lie in the positive-positive quadrant.

Benchmark gain (ΔA) and correct-image dependence (ΔG). Table 1 reports the per-seed quantities. ΔA is positive for every seed on both backbones (7B fresh-seed mean $+0.0506 \pm 0.0046$; 2B fresh-seed mean $+0.0316 \pm 0.0009$). The normal-minus-shuffle gap G widens under tuning in every family, and the zero-shot-relative change ΔG is positive for every seed (7B fresh-seed mean $+0.0440 \pm 0.0054$; 2B fresh-seed mean $+0.0316 \pm 0.0026$). Shuffle accuracy itself stays flat (e.g., 2B zero-shot 0.4692 vs. tuned 0.4674–0.4729), so the gain is in using the correct image, not in memorizing image-independent answers.

Semantic pair consistency (ΔC , relcomp). Table 2 shows relcomp C_{pair} . Fresh seeds cluster tightly around seed-0 in both families (7B: 0.6547–0.6652 vs. 0.6772; 2B: 0.4985–0.5105 vs. 0.5015); the audit confirms every fresh seed is within 0.05 of the legacy General checkpoint. The facing-antonym diagnostic (facingcomp, $n = 103$) shows the same pattern of tight seed clustering (2B seed-0 0.3469; fresh seeds 0.3429–0.3714).

family	checkpoint	C_{pair} (relcomp)
Qwen2-VL-7B	seed-0	0.6772
Qwen2-VL-7B	seedA	0.6637
Qwen2-VL-7B	seedB	0.6547
Qwen2-VL-7B	seedC	0.6652
SmolVLM2-2B	seed-0	0.5015
SmolVLM2-2B	seedA	0.4985
SmolVLM2-2B	seedB	0.5045
SmolVLM2-2B	seedC	0.5105

Table 2. Relcomp pair consistency ($n = 666$). Fresh seeds stay within 0.05 of the legacy General checkpoint in both families.

family	checkpoint	$A_{\text{transform}}$	C_{pair}	both_correct
Qwen2-VL-7B	zero-shot	0.6367	0.6163	0.5388
Qwen2-VL-7B	seed-0	0.6571	0.6857	0.5959
Qwen2-VL-7B	seedA	0.6490	0.6490	0.5796
Qwen2-VL-7B	seedB	0.6571	0.6898	0.6041
Qwen2-VL-7B	seedC	0.6449	0.6653	0.5837
SmolVLM2-2B	zero-shot	0.4980	0.3184	0.2531
SmolVLM2-2B	seed-0	0.5224	0.3469	0.2980
SmolVLM2-2B	seedA	0.5469	0.3429	0.3061
SmolVLM2-2B	seedB	0.5469	0.3633	0.3143
SmolVLM2-2B	seedC	0.5469	0.3714	0.3224

Table 3. Tier-C hflip_flip ($n = 245$). $A_{\text{transform}}$ is transformed-answer accuracy; C_{pair} is the answer-update rate $P(\text{mirrored} \neq \text{normal})$; both_correct is joint correctness. Three separate quantities, never collapsed.

Transformation behavior under global reflection (Tier C). Table 3 reports the three Tier-C quantities separately. Under global horizontal reflection (language fixed, left/right truth flips):

- **Transformed accuracy** ($A_{\text{transform}}$): 2B rises from 0.4980 (zero-shot) to 0.5224 (seed-0) and 0.5469 for each fresh seed; 7B 0.6367 $\rightarrow$ 0.6571 with fresh seeds 0.6490–0.6571.
- **Answer-update rate** (C_{pair}): 2B 0.3184 (zero-shot) $\rightarrow$ 0.3469 (seed-0), with all three fresh seeds higher than zero-shot (0.3429/0.3633/0.3714); 7B 0.6163 $\rightarrow$ 0.6857, fresh seeds 0.6490/0.6898/0.6653.
- **Joint correctness** (both_correct): 2B 0.2531 $\rightarrow$ 0.2980 (seed-0) $\rightarrow$ 0.3061/0.3143/0.3224.

The audit confirms all fresh 2B C_{pair} values exceed the 2B zero-shot value, and all fresh-seed C_{pair} values (both families) are within 0.05 of the legacy General checkpoint.

Contemporary-backbone extension (Qwen3-VL-8B, post-confirmatory). Table 1 includes the Qwen3-VL-8B row: $\Delta A + 0.0323$, $\Delta G + 0.0355$, and hflip_flip transformed accuracy $+0.0449$ (0.6571 $\rightarrow$ 0.7020). This extension is explicitly exploratory: it computed only ΔA , ΔG , and

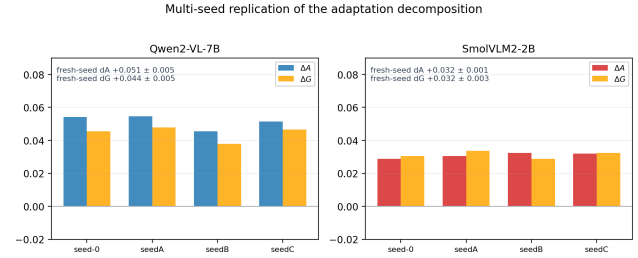


Figure 2. Multi-seed replication: per-seed ΔA and ΔG for both confirmatory families, with fresh-seed mean $\pm$ SD annotations.

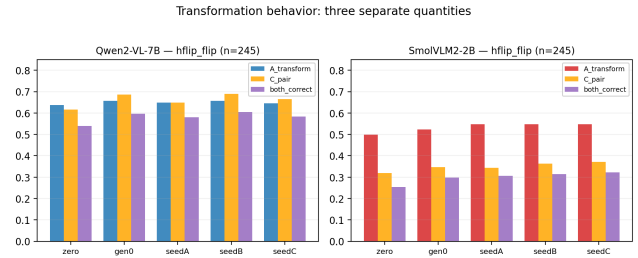


Figure 3. Transformation behavior: $A_{\text{transform}}$, C_{pair} , and both_correct per checkpoint, hflip_flip ($n = 245$), both families.

$A_{\text{transform}}$: C_{pair} was not computed for it, and no response-law-compliance claim is made. It is labeled post-confirmatory external validation, not preregistered confirmatory evidence.

Claim-level summary (all audited).

- $\Delta A > 0$ for all fresh seeds, both confirmatory backbones: PASS.
- $\Delta G > 0$ for all fresh seeds, both confirmatory backbones: PASS.
- Fresh-seed C_{pair} within 0.05 of legacy General (7B and 2B): PASS.
- All fresh 2B hflip_flip $C_{\text{pair}} > 2B$ zero-shot: PASS.
- Qwen3-VL-8B supports only the stated $\Delta A/\Delta G/A_{\text{transform}}$ quantities: PASS.

5. Discussion and Limitations

What the decomposition shows. The headline result is that ordinary spatial fine-tuning is not a scalar capability update. Benchmark accuracy, correct-image dependence, and semantic pair consistency move together in direction (all positive for ΔA and ΔG ; ΔC stable) but they are distinct quantities: shuffle accuracy stays flat while the gap widens; answer-update rates improve while transformed accuracy moves differently; and the three Tier-C quantities never collapse into one number. A model can get the mirrored item right without updating its answer relative to its own original prediction, and vice versa; reporting them separately is essential.

Seed robustness. Every component replicates across three fresh training draws per family. The fresh-seed standard deviations are small (7B ΔA SD 0.0046; 2B ΔA SD 0.0009), which closes the single-checkpoint-anecdote criticism that a training effect could be a lucky draw.

Relationship to prior work. We do not claim the normal-minus-shuffle gap as novel: VRS in Beyond Accuracy [6] is the same construction. We do not claim to reproduce VisualFLIP [7]: our reflection is a global image intervention, and we state the difference. Our novelty is the joint, seed-robust, cross-backbone decomposition of ordinary fine-tuning into ΔA , ΔG , ΔC , and Tier-C behavior.

Limitations. (1) Two backbone families (7B and 2B) plus one contemporary extension; the extension computed only $\Delta A/\Delta G/A_{\text{transform}}$ and is exploratory. (2) The Tier-C intervention is a global horizontal reflection; VisualFLIP-style minimal local edits were not run because the official dataset is not yet public. (3) All comparisons are within-backbone with identical preprocessing; we do not claim resolution or architecture independence. (4) The shuffle condition tests dependence on the *correct image*, which we call correct-image dependence, not “causal grounding”; stronger causal language is reserved for genuine counterfactual evidence. (5) A text-only control adapter was considered but not run; it remains future work. (6) Seeds are independent draws; we do not claim a “monotonic” seed trend anywhere.

6. Conclusion

Ordinary spatial LoRA fine-tuning on VSR produces a reproducible vector of capability changes, not a single benchmark gain. Across three fresh training seeds and two backbone families, ΔA (benchmark accuracy) and ΔG (correct-image dependence) are positive for every seed, semantic pair consistency ΔC stays within 0.05 of the legacy General checkpoint, and Tier-C transformation behavior improves answer-update rates while remaining a set of three separately reported quantities. A contemporary Qwen3-VL-8B extension reproduces the primary $\Delta A/\Delta G$ pattern and the transformed-accuracy gain. Every number was independently recomputed from raw artifacts and the frozen protocol; the full claim hierarchy, numerical audit, literature audit, figures, and tables are archived in the repository. We recommend that spatial fine-tuning be evaluated and reported as this joint vector—accuracy, correct-image dependence, semantic consistency, and transformation behavior—rather than collapsed into a single accuracy number.

References

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